General Mathematics — Exercise Sheet 21

Determinants and invertibility

Variant: Statements

Exercise 1 (Two rows changed at once). A real 3×3 matrix A has rows r_1, r_2, r_3 and determinant 5. For $c \in \mathbb{R}$, let B_c have rows $r_1 + 2r_2$, $r_1 + cr_2$ and $-3r_3$, in that order. Compute $\det B_c$ using row linearity and the vanishing of repeated-row terms. Find all c for which B_c is singular. Keep the other rows fixed at each application of linearity.

Exercise 2 (Three exceptional parameters). For $t \in \mathbb{R}$, let

$$A_t = \begin{pmatrix} t & 1 & 0 \\ 1 & t & 1 \\ 0 & 1 & t \end{pmatrix}.$$

Use elimination with determinant factors recorded to compute $\det A_t$ for every t. Classify invertibility. For each singular parameter, give a basis of $\ker A_t$ and its dimension under Lecture 18's convention. Avoid dividing by t, since $t = 0$ is included.

Exercise 3 (What determinant data determine). Real 3×3 matrices A and B satisfy $\det A = -2$ and $\det B = 3$. Compute $\det(2A^{-1}B^2)$ and $\det(B^{-1}AB)$, explaining why both inverses exist and how many scalar factors the whole-matrix scaling contributes. To test whether those two determinant data also determine $\det(A + B)$, compare the concrete matrices

$$A_0 = \operatorname{diag}(-2, 1, 1), \quad B_1 = \operatorname{diag}(3, 1, 1), \quad B_2 = \operatorname{diag}(1, 3, 1).$$

Here diag denotes the diagonal matrix with the listed diagonal entries and all other entries zero. Compute $\det(A_0 + B_1)$ and $\det(A_0 + B_2)$ and state your conclusion.

Exercise 4 (A zero diagonal entry before elimination). Compute the determinant of the real matrix

$$D = \begin{pmatrix} 1 & 2 & 0 & 3 & 1 \\ 2 & 5 & 0 & 6 & 2 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 3 & 1 & -1 \\ 0 & 0 & 6 & 0 & 1 \end{pmatrix}$$

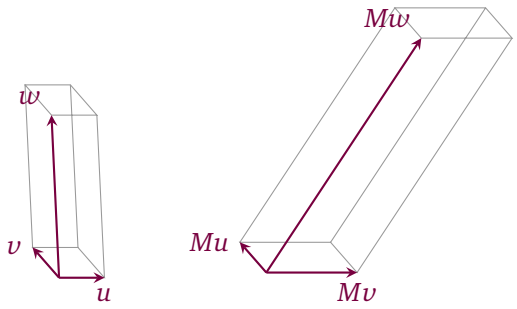
by Gaussian elimination, recording swaps and scalings. Explain why the initial zero in position $(3, 3)$ does not itself prove that the matrix is singular. State whether the resulting system $Dx = b$ has a unique solution for every $b \in \mathbb{R}^5$.

Exercise 5 (Signed volumes under a real map). Let a real linear map have matrix

$$M = \begin{pmatrix} 0 & 2 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

An ordered parallelepiped has side columns $u = (1,0,0)$, $v = (0,1,0)$ and $w = (1,2,3)$. Find its signed and ordinary volumes, then find the image columns Mu, Mv and Mw. Use determinant multiplicativity to find the image's signed and ordinary volumes and determine whether the orientation reverses. Find the signed volume obtained by applying M^{-1} to the original three columns, and the signed volume obtained by applying M twice to those original columns. Use Lecture 13's scalar-triple-product meaning and Lecture 21's product and inverse laws.

The drawings use the same projection and coordinate scales for the original and image parallelepipeds.



Exercise 6 (A Vandermonde determinant and three prescribed values). For $a, b, c \in \mathbb{R}$, compute the determinant

$$V = \begin{pmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{pmatrix}$$

by row reduction and factoring row scalars. Obtain a formula valid also when two nodes coincide, without dividing by any difference of nodes. Determine exactly when a polynomial $q(x) = \alpha + \beta x + \gamma x^2$ of degree at most 2 is uniquely determined by arbitrary prescribed values $q(a) = y_a$, $q(b) = y_b$ and $q(c) = y_c$. Use Lecture 20's all-right-sides inversion criterion.